NEHA DABEERU

2403860822 | Raleigh, NC | nehadabeeru@outlook.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Python Backend Software Engineer with 4+ years of experience building scalable APIs, microservices, event-driven systems & distributed workflows on AWS. Experienced in GraphQL/REST API design, asynchronous processing, PostgreSQL, DynamoDB & cloud-native architectures. Proven track record improving latency, reliability & production resilience through idempotency, automated testing, observability & failure recovery.

PROFESSIONAL EXPERIENCE

Software Engineer, Data & AI Platforms, Merck Sharp & Dohme (New Jersey, USA) *Jan 2025 – Present*

- Built AppSync GraphQL APIs with Lambda and VTL resolvers for encoding, approval, audit, & external-partner workflows on a **3M+** req/day platform.
- Eliminated AppSync **30s** timeouts & cut Databricks latency from **60s to 2s** with async orchestration, caching, connection pooling & request batching.
- Led E2E delivery from requirements & system design through estimation, implementation, UAT & prod sign-off, with engineering, QA, & stakeholders.
- Engineered versioned specification platform with RBAC, audit trails & workflow states, reducing manual handoffs and approval turnaround by **60%**.
- Drove GenAI auto-coding with BM25, Bedrock embeddings & neural k-NN retrieval, at **99.52%** top-match accuracy via weighted score normalization.
- Automated event-driven ingestion with Step Functions, Databricks, S3 & OpenSearch, enabling **5–6** hour incremental syncs and nightly full loads.
- Raised backend test coverage to **90%+** with pytest, moto & AWS integration tests, preventing regressions across workflow & data-integrity paths.
- Reduced repeat prod incidents through on-call triage, root-cause analysis, postmortems & permanent fixes across APIs, workflows & data pipelines.

Software Engineer II, Bank of America (North Carolina, USA) *Jun 2024 – Aug 2024*

- Modernized legacy codebase using AI-assisted refactoring, consolidating duplicate logic into reusable modules & expanding automated test coverage.
- Built checksum reconciliation framework to detect schema drift, missing records, and silent data mismatches for Cards and Payments migration.
- Automated source-to-target validation for referential integrity, data types, record counts, and field values, reducing pre-cutover verification effort.
- Generated exception reports covering mismatches, affected records & failure summaries, speeding defect triage, remediation & cutover decisions.

Software Engineer I, Bank of America (Tamil Nadu, India) *Jun 2020 – Jan 2023*

- Delivered Python backend services integrating trading systems & SQL databases for market and credit-risk data calculations & regulatory reporting.
- Improved regulatory reporting accuracy to 99.5% with automated reconciliation detecting missing records, balance mismatches & data-quality issues.
- Reduced risk-batch latency **40%** by parallelizing Python workloads & tuning complex queries, indexes & database access across high-volume datasets.
- Deployed rule-based SQL translation microservice parsing legacy syntax and converting to target dialect, reducing manual migration effort by **80%**.
- Increased workflow resilience with idempotent execution & checkpoint-based recovery, resuming failed workloads from the last successful stage.

Software Engineer Intern, Rashtriya Ispat Nigam Limited (Andhra Pradesh, India) *Jul 2017 – Sep 2017*

- Developed Python services for ERP-integrated workflows, automating validation and processing of **10K+** manufacturing and operational records/day.
- Created database-driven web application with reusable business logic and server-side validations, supporting 500+ users across enterprise workflows.

EDUCATION

M.S., Data Analytics, San Jose State University, California Jan 2023 – Dec 2024

Relevant Coursework: Cloud Computing, Database Systems, Data Engineering, Machine Learning, Big Data Analytics, Statistical Modeling

B.E., Computer Science and Engineering, SRM University, India June 2016 – May 2020

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Distributed Systems, Web Technologies

SELECTED ENGINEERING PROJECTS

DataFlo & DQ Checker — GenAI Clinical Data Platform:

- Cut ETL onboarding from 8–10 weeks to 1 week with GenAI auto-mapping, semantic retrieval, prompt-based reranking & data standardization.
- Developed multi-model RAG across Bedrock Haiku & OpenAI with confidence scoring and fallback chains, reaching **90%+** auto-mapping accuracy.
- Eliminated manual authoring for **100+** data-quality checks with LLM natural-language rule engine with validation, real-time alerts, & audit trails.

MedPredict — Hospital Readmission Prediction Platform:

- Reached **92%+** readmission accuracy with transformer embeddings, FAISS retrieval, and PyTorch classification across **40K+** MIMIC-III records.
- Cut inference latency to **2** secs by deploying containerized FastAPI service on AWS with async request handling and model-serving optimizations.

TECHNICAL SKILLS

Languages & Backend: Python, Java, SQL, JavaScript, TypeScript, REST APIs, GraphQL, FastAPI, AWS AppSync, API Gateway, AWS Lambda, VTL, Microservices, API Design, Authentication, Authorization

Cloud, Distributed Systems & Databases: AWS, Step Functions, EventBridge, SQS, Event-Driven Architecture, Asynchronous Processing, Workflow Orchestration, Idempotency, Retry Strategies, Failure Recovery, Concurrency Control, PostgreSQL, MySQL, DynamoDB, Amazon OpenSearch, pgvector

DevOps, Testing, Observability & AI: pytest, moto, Docker, Terraform, GitHub Actions, CI/CD, Git, CloudWatch, X-Ray, Structured Logging, Distributed Tracing, AWS Bedrock, RAG, Vector Search, Apache Spark, PySpark, Databricks

CERTIFICATION

AWS Certified Data Engineer – Associate, Amazon Web Services May 2025 – May 2028